MATHEMATICS - FORMULA EXTRACTION TEST

1) Evaluate:

$$\int_{1}^{\square} (x^2 + \sqrt{(1 + x\square)}) / (1 + x^2)\, dx$$

2) Differential equation:

$$\partial^2 u/\partial x^2 + \partial^2 u/\partial y^2 = \lambda u$$

3) Quadratic formula:

$$x = [-b \pm \sqrt{(b^2 - 4ac)}] / (2a)$$

4) Matrix:

$$A = \begin{bmatrix} a\square\square & a\square\square & a\square\square \\ a\square\square & a\square\square & a\square\square \\ a\square\square & a\square\square & a\square\square \end{bmatrix}$$

5) Summation:

$$\sum_{\square}^{\square\square} k^2 = n(n + 1)(2n + 1) / 6$$

6) Probability:

$$P(A \cap B) = P(A)P(B \mid A)$$

7) Trigonometry:

$$\sin^2\theta + \cos^2\theta = 1$$

8) Harder expression:

$$f(x) = (e^{(-x^2)}/\sqrt{\pi}) \int_{\infty}^{\square} e^{(-t^2)}\, dt$$

OCR TEST: preserve equation structure, Greek symbols, powers, roots and limits.